Customer Segmentation Using K-Means and Hierarchical Clustering for Purchasing-Behavior Analysis

Research Paper | *Nassau Candy Distributor Dataset*

Abstract

This study examines unsupervised customer segmentation using a supplied candy-distribution transaction dataset. The source file contains 10,194 transaction rows, 5,044 distinct customers, 8,549 distinct orders, and 15 distinct products. Customer-level features were created from sales, gross profit, units, order frequency, product variety, and average purchase measures. Standardized numeric features were evaluated with K-Means and Ward-linkage agglomerative hierarchical clustering for cluster counts from two to ten. For the four-cluster configuration, the sampled silhouette score was 0.4639 for K-Means and 0.4865 for hierarchical clustering. The K-Means four-cluster solution was retained as a practical interpretation candidate because its smallest cluster was larger and its size distribution was less concentrated than the hierarchical four-cluster solution; this is a usability choice rather than a claim of metric superiority. The resulting K-Means segments differ in spending, order frequency, and product variety. The analysis is exploratory and does not establish causal effects or prove that proposed marketing actions will improve outcomes. A significant data-quality limitation was identified: recorded ship dates occur far later than order dates, yielding delays of 904 to 1,642 days; shipping-date-based conclusions were therefore excluded.

Keywords—customer segmentation, K-Means, hierarchical clustering, unsupervised learning, exploratory data analysis, purchasing behavior, retail analytics.

I. Introduction

Customer segmentation groups customers with similar observed characteristics so that organizations can summarize heterogeneous purchasing behavior and investigate differentiated engagement strategies. In transaction-based retail analytics, segmentation can be constructed from customer-level spending, purchase frequency, product variety, and profit contribution. Unsupervised clustering is appropriate when no pre-existing target label defines the customer groups, but the resulting clusters must be interpreted using the original business measures rather than treated as ground-truth customer types.

This project applies two clustering methods to a supplied Nassau Candy Distributor dataset. The original project brief had referred to real-estate investment profiling, but the provided CSV contains candy-distribution transactions, not property purchases, financing indicators, buyer demographics, or real-estate investment outcomes. Consequently, the present study is explicitly scoped to purchasing-behavior segmentation in the supplied dataset.

Research objectives

- Inspect and summarize the supplied transaction dataset and document data-quality limitations.
- Aggregate transactions into customer-level features suitable for clustering.
- Compare K-Means and Ward-linkage hierarchical clustering across candidate cluster counts.
- Describe the four-cluster K-Means solution using interpretable business measures.
- Translate observed patterns into cautious, testable recommendations and a dashboard specification.

II. Dataset and Exploratory Data Analysis

The supplied CSV has 10,194 rows and 18 columns. It contains order identifiers and dates, shipping mode, customer ID, geographic fields, product identifiers and names, sales, units, gross profit, and cost. The dataset includes 5,044 distinct customers, 8,549 distinct orders, and 15 distinct products. No exact duplicate transaction rows or duplicate Row ID values were found. The file had no missing values across its 18 columns during the validation performed for this submission.

Dataset measure	Verified value
Transaction rows	10,194
Unique customers	5,044
Unique orders	8,549
Unique products	15
Total sales	$141,783.63
Total cost	$48,340.83
Total gross profit	$93,442.80
Units sold	38,654
Aggregate gross profit margin	65.91%

The aggregate gross profit margin is calculated as total gross profit divided by total sales. It is not a net profit margin and should not be interpreted as such. Sales, cost, and gross profit are mathematically related, so strong correlations among these columns are expected and should not be interpreted as independent evidence of causal drivers.

Data-quality limitation: dates

Order dates parsed from the provided day-month-year strings range from 2024-01-02 to 2025-12-31. Ship dates range from 2026-06-30 to 2030-06-28. All 10,194 rows have a ship date later than the order date, with calculated differences ranging from 904 to 1642 days. These intervals are suspicious for ordinary shipping and could reflect source-data issues or a date transformation. They were not corrected because the intended values are not known; shipping-delay analysis is excluded pending provider verification.

III. Methodology

A. Customer-level aggregation and feature engineering

The transaction table was aggregated by Customer ID. The selected clustering inputs were total sales, total gross profit, total units, distinct order count, distinct product count, average sales per order, and average units per order. Customer ID was retained only as an identifier and was not used as a distance feature. Transaction count was not included in the initial feature vector because it overlaps with order activity. Gross profit margin and categorical preferences were retained for interpretation but not included in the seven-feature numeric clustering matrix.

B. Standardization

The seven numerical features were standardized using StandardScaler so that features with larger raw scales would not dominate Euclidean-distance-based clustering solely because of their units. The saved scaled file contains 5,044 rows and seven features. Identical standardized feature vectors can occur for different customers; repeated feature rows do not imply duplicate customer identities.

C. Clustering algorithms

K-Means partitions observations into K groups by minimizing within-cluster sum of squared distances to cluster centroids. The experiment used random_state=42 and n_init=10. Agglomerative hierarchical clustering used Ward linkage, which merges groups to minimize increases in within-cluster variance. Both methods were evaluated for K=2 through K=10.

D. Evaluation metrics

The silhouette coefficient summarizes cohesion and separation; larger values generally indicate more distinct clusters. The Davies–Bouldin index measures average similarity between each cluster and its most similar alternative; lower values are preferred. The Calinski–Harabasz index compares between-cluster dispersion with within-cluster dispersion; larger values are generally preferred. The supplied evaluation workflow used a 2,000-customer sample for silhouette scores, while Davies–Bouldin and Calinski–Harabasz scores were computed on the full dataset. Consequently, silhouette values should be described as sample estimates rather than full-data exact values.

IV. Results and Discussion

A. Model comparison

Metric at K=4	K-Means	Hierarchical
Silhouette (sample estimate)	0.4639	0.4865
Davies–Bouldin (lower is better)	0.8858	0.8775
Calinski–Harabasz (higher is better)	3327.17	2453.91
Smallest cluster	339	84
Largest cluster	2413	3720

At K=4, hierarchical clustering has a higher sampled silhouette estimate (0.4865) and a lower Davies–Bouldin index (0.8775) than K-Means. K-Means has a higher Calinski–Harabasz score (3327.17 versus 2453.91) and a less extreme cluster-size distribution (339–2,413 customers versus 84–3,720). The K-Means K=4 solution was carried forward for interpretability and operational practicality, not because it dominates every metric.

B. K-Means segment profiles

Cluster	Customers	Customer share	Avg. sales/customer	Total sales	Sales share
0	2,413	47.84%	$10.97	$26,463.41	18.66%
1	339	6.72%	$101.42	$34,382.51	24.25%
2	896	17.76%	$35.08	$31,428.01	22.17%
3	1,396	27.68%	$35.47	$49,509.70	34.92%

Cluster 0 is the largest group (2,413 customers; 47.84%) and has mean customer sales of $10.97. Cluster 1 contains 339 customers (6.72%) with mean sales of $101.42 and contributes 24.25% of total sales. Cluster 2 contains 896 customers (17.76%) with mean sales of $35.08 and contributes 22.17% of sales. Cluster 3 contains 1,396 customers (27.68%) with mean sales of $35.47 and contributes 34.92% of sales. Cluster 3's sales contribution is the largest among the four groups. These descriptions are based on the generated cluster assignments and descriptive averages; they do not establish customer motivations or future behavior.

C. Selecting K and interpreting metrics

Across the tested configurations, K-Means at K=2 had the highest recorded silhouette estimate (0.5192) and the highest Calinski–Harabasz score (3703.86). Hierarchical clustering at K=10 had the lowest Davies–Bouldin index (0.7984), but its smallest cluster contained only 5 customers. K-Means at K=10 had a Davies–Bouldin index of 0.8019 and a smallest cluster of 14 customers. These results illustrate a trade-off between compactness/separation and operational interpretability. Four clusters were retained as a practical working choice for this submission; this choice should be revisited if stakeholders prefer fewer, simpler segments or if further stability and validation analysis becomes available.

V. Recommendations and Limitations

A. Recommendations

- Treat the four groups as exploratory customer profiles and avoid assigning psychological or demographic labels that are not supported by the data.
- Investigate low-cost retention and engagement experiments for the high-sales Cluster 1, because it represents 6.72% of customers and 24.25% of recorded sales.
- Evaluate reorder reminders for customers with infrequent purchasing, but use an experiment or holdout group to test whether reminders change repeat purchasing.
- Explore product bundles or cross-selling for customers with broader product variety, using division and product preference results as descriptive signals.
- Monitor both sales and gross profit when evaluating interventions; higher sales alone do not guarantee higher profit or better customer outcomes.

- Validate source dates, definitions of cost and gross profit, and the business meaning of customer/order identifiers before using the dashboard for operational decisions.

B. Limitations

- The dataset represents candy distribution transactions and cannot support claims about real-estate investment, property buyers, income, financing, or demographics.
- Ship dates are inconsistent with order dates and were not repaired; shipping-delay conclusions are not reported.
- Clustering is unsupervised: there are no ground-truth segment labels against which to measure classification accuracy.
- Cluster labels are arbitrary, and cluster assignments may vary with feature choices, scaling, algorithm settings, or data changes.
- Silhouette scores were estimated on a sample of 2,000 observations in the evaluation output.
- The analysis is descriptive and observational. It does not prove that recommended marketing actions will cause increased sales or profit.
- The dataset's time range and representativeness are limited to the supplied file; external generalization has not been tested.

VI. Conclusion and Future Work

This study demonstrates a reproducible workflow for customer-level feature construction, standardization, unsupervised clustering, model comparison, and segment interpretation using the supplied candy distributor transactions. The four-cluster K-Means solution provides an interpretable starting point: customer groups differ in average spending, frequency, and contribution to recorded sales. Hierarchical clustering provides a useful comparison and performs better on some metrics at K=4, while K-Means produces a less concentrated size distribution. The results should be treated as exploratory until the date anomaly is resolved, segmentation stability is tested, and business experiments validate the recommendations. Future work may include robust scaling or log transformation for skewed spending, repeated-seed stability checks, holdout-period validation, and controlled evaluation of targeted engagement strategies.